K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

School of Engineering & Technology

OPERATING SYSTEM

LAB ASSINGMENT 3

Program: BCA (AI & DS)

Course Code – ENCA252 (2026-2027)

Submitted By

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BCA (AI & DS)

Submitted To

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Task 1: Input and Page Reference String

Task 2: FIFO (First-In-First-Out)

```
Linux [Running] - Oracle VirtualBox
File    Machine    View    Input    Devices    Help

rakesh@rakesh-VirtualBox:~$ cd ~/Documents
rakesh@rakesh-VirtualBox:~/Documents$ nano page_replacement.py
rakesh@rakesh-VirtualBox:~/Documents$ nano page_replacement.py
rakesh@rakesh-VirtualBox:~/Documents$ python3 page_replacement.py

============================================================
      PAGE REPLACEMENT ALGORITHM SIMULATOR
============================================================

Enter number of frames: 3
Enter page reference string (space-separated): 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

Frames        : 3
Reference Str : [7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1]
Total Pages   : 20

────────────────────────────────────────────────────────────
  FIFO  (First-In First-Out)
────────────────────────────────────────────────────────────
  Page | Frames          | Status
──────────────────────────
     7 |   7             | FAULT
     0 |   7    0        | FAULT
     1 |   7    0    1   | FAULT
     2 |   2    0    1   | FAULT
     0 |   2    0    1   | HIT
     3 |   2    3    1   | FAULT
     0 |   2    3    0   | FAULT
     4 |   4    3    0   | FAULT
     2 |   4    2    0   | FAULT
     3 |   4    2    3   | FAULT
     0 |   0    2    3   | FAULT
     3 |   0    2    3   | HIT
     2 |   0    2    3   | HIT
     1 |   0    1    3   | FAULT
     2 |   0    1    2   | FAULT
     0 |   0    1    2   | HIT
     1 |   0    1    2   | HIT
     7 |   7    1    2   | FAULT
     0 |   7    0    2   | FAULT
     1 |   7    0    1   | FAULT

  ▸ Total Page Faults: 15

────────────────────────────────────────────────────────────
  LRU   (Least Recently Used)
────────────────────────────────────────────────────────────
  Page | Frames          | Status
──────────────────────────
     7 |   7             | FAULT
```

Task 3: LRU (Least Recently Used)

Task 4: Optimal Page Replacement

Linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
LRU   (Least Recently Used)
Page | Frames          | Status
   7 |   7             | FAULT
   0 |   7    0        | FAULT
   1 |   7    0    1   | FAULT
   2 |   0    1    2   | FAULT
   0 |   1    2    0   | HIT
   3 |   2    0    3   | FAULT
   0 |   2    3    0   | HIT
   4 |   3    0    4   | FAULT
   2 |   0    4    2   | FAULT
   3 |   4    2    3   | FAULT
   0 |   2    3    0   | FAULT
   3 |   2    0    3   | HIT
   2 |   0    3    2   | HIT
   1 |   3    2    1   | FAULT
   2 |   3    1    2   | HIT
   0 |   1    2    0   | FAULT
   1 |   2    0    1   | HIT
   7 |   0    1    7   | FAULT
   0 |   1    7    0   | HIT
   1 |   7    0    1   | HIT

➤ Total Page Faults: 12

Optimal Page Replacement
Page | Frames          | Status
   7 |   7             | FAULT
   0 |   7    0        | FAULT
   1 |   7    0    1   | FAULT
   2 |   2    0    1   | FAULT
   0 |   2    0    1   | HIT
   3 |   2    0    3   | FAULT
   0 |   2    0    3   | HIT
   4 |   2    4    3   | FAULT
   2 |   2    4    3   | HIT
   3 |   2    4    3   | HIT
   0 |   2    0    3   | FAULT
   3 |   2    0    3   | HIT
   2 |   2    0    3   | HIT
   1 |   2    0    1   | FAULT
   2 |   2    0    1   | HIT
   0 |   2    0    1   | HIT
   1 |   2    0    1   | HIT
   7 |   7    0    1   | FAULT
```

Task 5: MRU (Most Recently Used)

```
rakesh@rakesh-VirtualBo

MRU   (Most Recently Used)
Page | Frames          | Status
   7 |   7            | FAULT
   0 |   7    0       | FAULT
   1 |   7    0    1  | FAULT
   2 |   7    0    2  | FAULT
   0 |   7    0    2  | HIT
   3 |   7    3    2  | FAULT
   0 |   7    0    2  | FAULT
   4 |   7    4    2  | FAULT
   2 |   7    4    2  | HIT
   3 |   7    4    3  | FAULT
   0 |   7    4    0  | FAULT
   3 |   7    4    3  | FAULT
   2 |   7    4    2  | FAULT
   1 |   7    4    1  | FAULT
   2 |   7    4    2  | FAULT
   0 |   7    4    0  | FAULT
   1 |   7    4    1  | FAULT
   7 |   7    4    1  | HIT
   0 |   0    4    1  | FAULT
   1 |   0    4    1  | HIT

 ▸ Total Page Faults: 16

Second Chance (Clock Algorithm)
Page | Frames          | Status
   7 |   7            | FAULT
   0 |   7    0       | FAULT
   1 |   7    0    1  | FAULT
   2 |   2    0    1  | FAULT
   0 |   2    0    1  | HIT
   3 |   2    0    3  | FAULT
   0 |   2    0    3  | HIT
   4 |   4    0    3  | FAULT
   2 |   4    2    3  | FAULT
   3 |   4    2    3  | HIT
   0 |   4    2    0  | FAULT
   3 |   3    2    0  | FAULT
   2 |   3    2    0  | HIT
   1 |   3    1    0  | FAULT
   2 |   3    1    2  | FAULT
   0 |   0    1    2  | FAULT
   1 |   0    1    2  | HIT
```

Task 6: Performance Comparison

Task 7: Result Analysis

Linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

rakesh@rakesh-VirtualBox: ~/Documents

```
============================================================
         PERFORMANCE COMPARISON
============================================================
Algorithm                 |  Page Faults | Rank
__________________________________________________________
Optimal                   |            9 | #1
LRU                       |           12 | #2
Second Chance             |           14 | #3
FIFO                      |           15 | #4
MRU                       |           16 | #5

  ✔ Best  : Optimal  (9 faults)
  ✘ Worst : MRU (16 faults)

============================================================
         RESULT ANALYSIS & CONCLUSION
============================================================

  [FIFO]  –  15 page faults
    Simple queue-based replacement. Easy to implement.
    Con: Can suffer Belady's Anomaly. Ignores usage frequency.

  [LRU]  –  12 page faults
    Exploits temporal locality. Good real-world performance.
    Con: Requires usage tracking overhead.

  [Optimal]  –  9 page faults
    Theoretically minimum faults. Perfect benchmark.
    Con: Requires future knowledge – NOT usable in practice.

  [MRU]  –  16 page faults
    Good for cyclic/sequential scan workloads.
    Con: Poor for general workloads. Often worst performer.

  [Second Chance]  –  14 page faults
    Practical FIFO improvement using reference bits.
    Pro: Approximates LRU with low overhead. Used in Linux kernel.

  CONCLUSION:
  • Optimal gives minimum faults (9) but is impractical.
  • LRU (12 faults) is the best practical algorithm.
  • Second Chance (14 faults) is a good low-overhead alternative.
  • MRU and FIFO generally perform worse for typical reference strings.

============================================================
  Simulation Complete!
============================================================

rakesh@rakesh-VirtualBox:~/Documents$
```